Supplementary information

Virtual bronchoscopic navigation versus non- virtual bronchoscopic navigation assisted bronchoscopy for the diagnosis of peripheral pulmonary lesions: A systematic review and meta-analysis

Mohan Giri, Anju Puri, Ting Wang, Guichuan Huang, Shuliang Guo*

Department of Respiratory and Critical Care Medicine, the First Affiliated Hospital of Chongqing Medical University, Chongqing, China

* Corresponding author: Shuliang Guo

Department of Respiratory and Critical Care Medicine, the First Affiliated Hospital of Chongqing Medical University, No 1 Youyi Road, Yuzhong, Chongqing 400016, China; E-mail: cyhxgsl@sina.com

Contents

Appendix S1- Electronic search in PubMed database

Figure S1. Forest plot comparing the complications between the VBNA vs NVBNA group.

Figure S2. Risk of bias summary

Figure S3. Risk of bias graph

Figure S4. Forest plot comparing the diagnostic yield according to the nature of lesion of VBNA vs NVBNA group. VBNA = virtual bronchoscopic navigation assisted; NVBNA = non-virtual bronchoscopic navigation.

Figure S5. Forest plot comparing the diagnostic yield according to the location of lesion in the lung lobe of VBNA vs NVBNA group.

Figure S6. Forest plot comparing the diagnostic yield according to the distance from the hilum of VBNA vs NVBNA group.

Figure S7. Forest plot comparing the diagnostic yield according to the bronchus sign of VBNA vs NVBNA group.

Table S1. Analysis of current several bronchoscopic technologies (BTs).

Appendix S1- Electronic search in PubMed database

Search ID	***Search Terms***
#1	("Virtual bronchoscopic navigation" or virtual bronchoscop* or "VBN")
#2	("Peripheral lung lesions" OR "Peripheral Pulmonary Lesions")
#3	("randomized controlled trial"[pt] OR "controlled clinical trial"[pt] OR randomized [tiab] OR randomly [tiab])
#4	(("Virtual bronchoscopic navigation" or virtual bronchoscop* or "VBN")) AND (("Peripheral lung lesions" OR "Peripheral Pulmonary Lesions"))
#5	#4 AND #3

Figure S1. Forest plot comparing the complications between the VBNA vs NVBNA group.

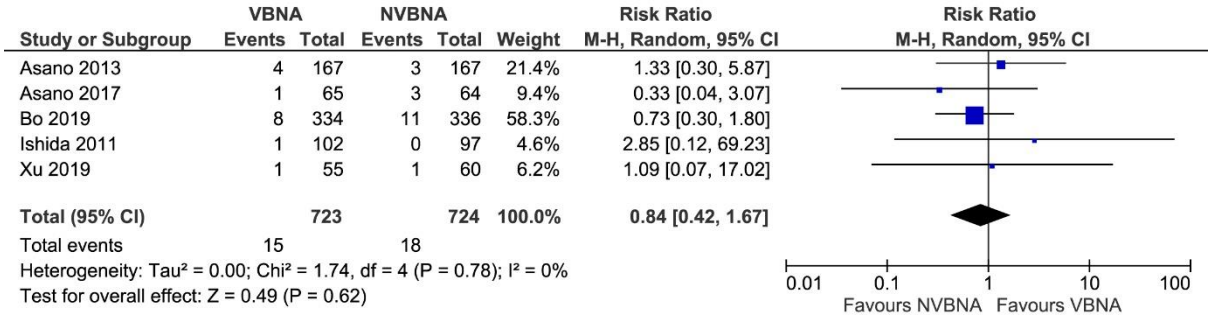
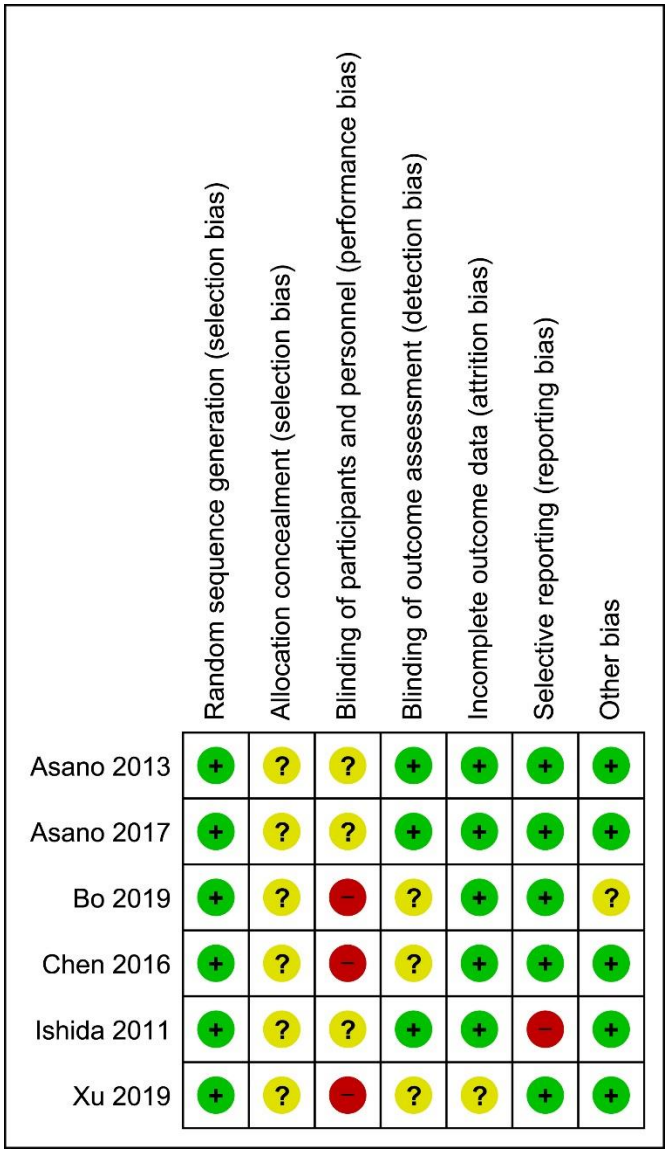
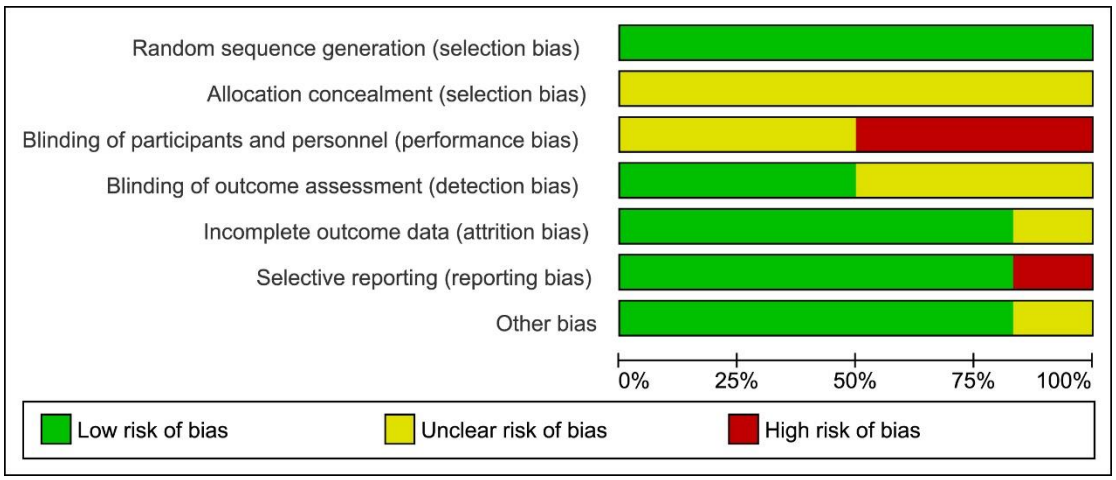


Figure S2. Risk of bias summary



Review authors' judgments for each risk of bias item for the included randomized controlled trials. All studies had adequate random sequence generation. Blinding of participants and personnel (3 studies), and selective reporting (1 study) were the main source of high risk of bias. Allocation concealment (6 studies), blinding of participants and personnel (3 studies), blinding of outcome assessment (3 studies), incomplete outcome data (1 study) and other bias (1 study) were the main source of unclear risk of bias.

Figure S3. Risk of bias graph



Review authors' judgments about each risk of bias item presented as percentages across all included randomized controlled studies.

Figure S4. Forest plot comparing the diagnostic yield according to the nature of lesion of VBNA vs NVBNA group.

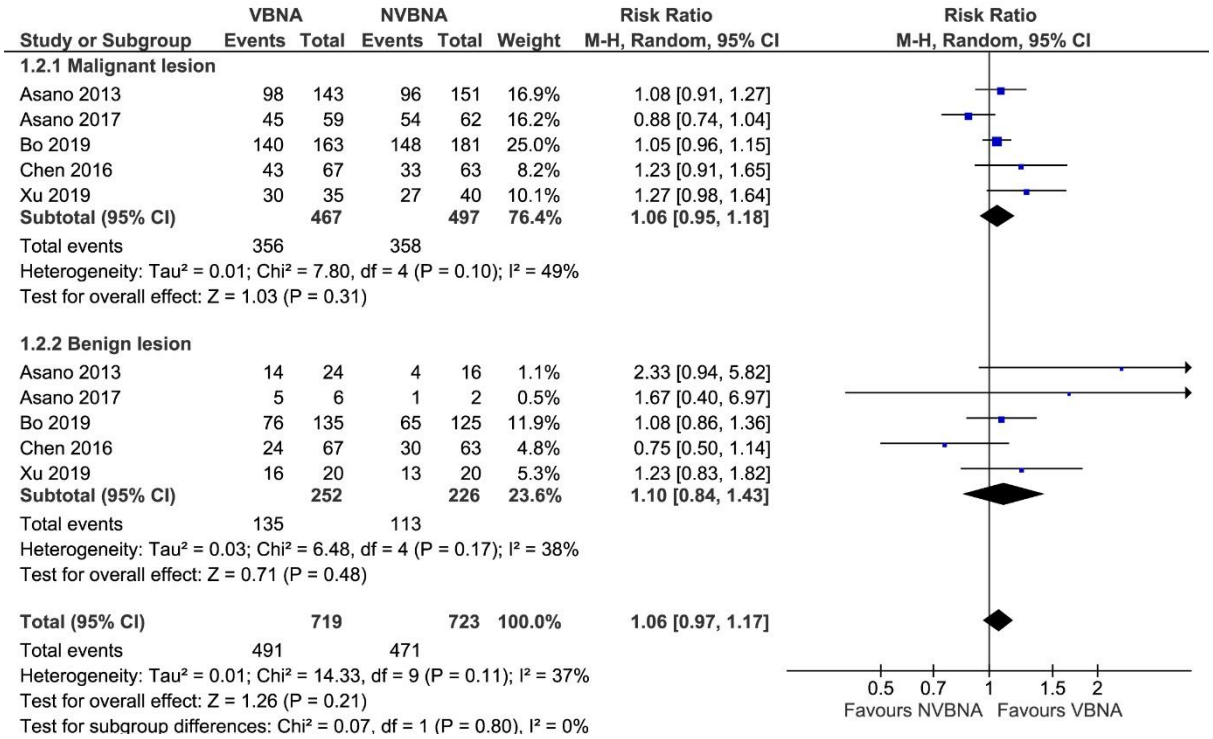


Figure S5. Forest plot comparing the diagnostic yield according to the location of lesion in the lung lobe of VBNA vs NVBNA group.

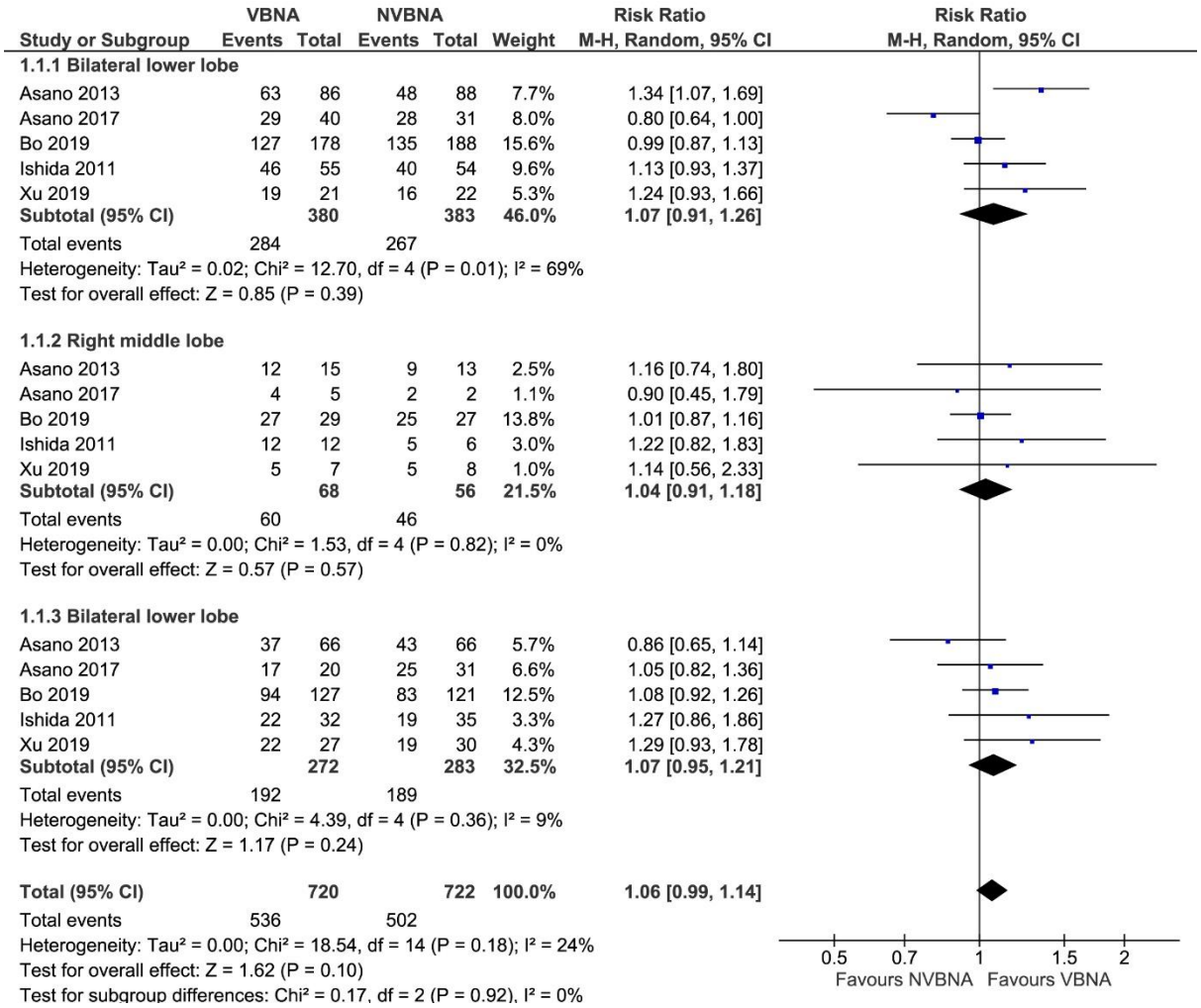


Figure S6. Forest plot comparing the diagnostic yield according to the distance from the hilum of VBNA vs NVBNA group.

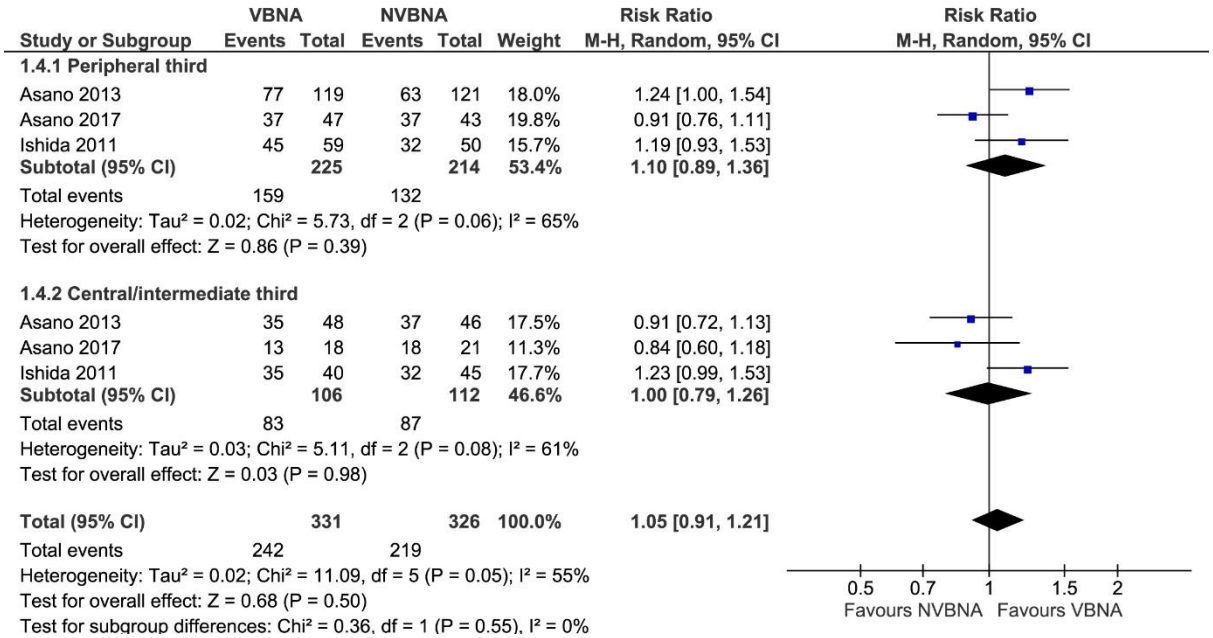


Figure S7. Forest plot comparing the diagnostic yield according to the bronchus sign of VBNA vs NVBNA group.

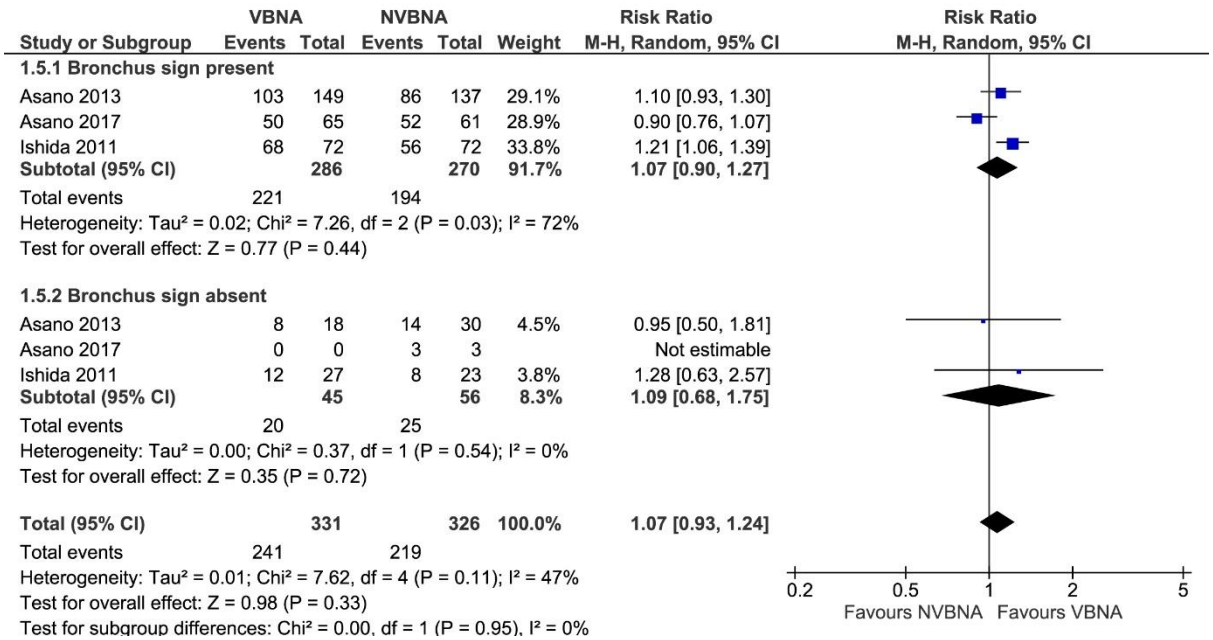


Table S1. Analysis of current several bronchoscopic technologies (BTs)

Bronchoscopic technologies (BTs)		Advantages	Disadvantages	Future work
Conventional BTs	Conventional transbronchial needle aspiration (C-TBNA)	1) Cost effective, safe, minimally invasive technique	1) Depends on operator experience 2) Inferior sensitivity 3) C-TBNA is performed without real-time ultrasound guidance, systemic training is must. 4) Significant cumulative radiation dose for the patient and the operator 5) Restricted range of motion	1) More Randomized controlled trials with conventional BTs combined with new BTs like VBN, ENB, and robotic bronchoscopy.
	CT-guided transthoracic needle aspiration (CT-TTNA)	1) High diagnostic yields even for small nodules 2) The standard diagnostic approach for PPLs	1) Complications like pneumothorax 2) Lacks the ability to perform simultaneous staging 3) Not suitable for lung lesions in the lower part of the chest, which moves with the diaphragm	
New BTs	Thin/Ultrathin bronchoscopes	1) Good accessibility to peripheral pulmonary lesions 2) High diagnostic yield	1) Can obtain specimens of only limited size	1) Randomized comparative studies with TTNA and conventional bronchoscopy are needed to clarify the position of VBN or ENB in the diagnostic methods for PPLs 2) Cost/benefit analysis should be carried out 3) Education programs and courses on VBN or ENB should be offered and performed to increase the skill level of operators.
	Radial probe endobronchial ultrasound (RP-EBUS)	1) Has a higher diagnostic yield and lower complication rate than conventional bronchoscopy	1) Higher costs, as well as timing and specific skills related to the ultrasound technology is required 2) Not available in several institutions	
	Virtual navigation bronchoscopy (VBN)	1) VBN has lower cost in comparison with ENB 2) VBN has the technical simplicity no specific training is required. 3) Not needing a sensor or a specific biopsy instrument 4) Complication rate is lower 5) High diagnostic yield for small lesion ≤2 cm	1) VBN cannot be a substitute for X-ray fluoroscopy 2) VBN cannot improve the diagnostic sensitivity 3) VBN guidance is limited by insufficient segmentation of the peripheral airways	
	Electromagnetic navigation	1) ENB has real-time navigation that subsequently directs	1) Requires expensive disposable	

	bronchoscopy (ENB)	the bronchoscopist to the lesion, making diagnosis more accurate	instruments for each patient. 2) ENB achieved a higher specificity, but is invasive and may cause pain 3) Requires expertise training	
	Robotic bronchoscopy	1) Could advance significantly further into the bronchial tree, particularly in difficult airways 2) More precise sample of the target lesion with less complications.	1) Expensive 2) Increased complexity in the operating room 3) Increased procedure time	
More novel BTs	Bronchoscopic transparenchymal nodule access (BTPNA)	1) BTPNA can detect PPLs regardless of CT bronchus sign 2) Less risk of adverse events.	1) Invasiveness technique. 2) The procedure may appear more complicated than a transthoracic CT-guided biopsy	1) Cost-effectiveness studies are needed. 2) More studies and clinical application is needed. 3) Further investigation on safety is necessary
	Transbronchial access tool (TBAT)	1) Promising for sampling lesions Without a bronchus sign. 2) Less risk of adverse events.	1) Invasiveness technique. 2) The procedure may appear more complicated than a transthoracic CT-guided biopsy.	